

Construction of Aamli Fala Bridge and Road

CD WORK DETILAD

S.N	CH	TYPE	Span Detail	Width
1	0/075	HPC	1 ROW 1000 MM	7.50
2	0/175	HPC	1 ROW 1000 MM	7.50
3	0/275	HPC	2 ROW 1000 MM	7.50
4	0/425	HPC	2 ROW 1000 MM	7.50
5	0/650	HPC	2 ROW 1000 MM	7.50
6	0/775	HPC	2 ROW 1000 MM	7.50
7	0/925	HPC	1 ROW 1000 MM	7.50
8	1/125	HPC VCW 60 Mtr	10 ROW 1000 MM	7.50
9	1/325	HPC	1 ROW 1000 MM	7.50
10	1/425	HPC	1 ROW 1000 MM	7.50
11	1/450	HPC	2 ROW 1000 MM	7.5
12	1/550	HPC	1 ROW 1000 MM	7.50
13	1/625	HPC	1 ROW 1000 MM	7.50
14	1/775	HPC	1 ROW 1000 MM	7.50
15	1/950	HPC	1 ROW 1000 MM	7.50
16	2/100	HPC	1 ROW 1000 MM	7.50
17	2/175	HPC	1 ROW 1000 MM	7.50
18	2/375	HPC	1 ROW 1000 MM	7.50
19	2/950	HPC	2 ROW 1000 MM	7.50
20	3/450	HPC	2 ROW 1000 MM	7.50
21	3/930	HPC	2 ROW 1000 MM	7.50

RETAINING WALL

1		R/wall 0/500 to 1/0	250.00 Mtr	
2		R/wall 1/0 to 1/500	240.00 Mtr	
3		R/wall 1/300 to 1/600	120.00 Mtr	
4		R/wall 1/850 to 2/00	80.00 Mtr	

DETAILED ESTIMATE

COST ESTIMATE FOR CONSTRUCTION OF CROSS DRAINAGE WORKS

(Culverts and Minor Bridges)

NAME OF ROAD :-

Construction of Aamli Fala Bridge and Road

Type of cross Drainage works : Pipe Culvert - 1000mm Dia. 1Row

M³ /Sec.

District : DUNGARPUR

: HSMB

1 Row 1000mm dia. 18 Width = 7.50

Sl.No.	Description of Item	No.	L (m)	B (m)	D (m)	Quantity	Qty 2Nos	Unit
1	2	3	4	5	6	7	8	9
1	Earthwork in excavation for B-1 foundation of structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. n ordinary soil [MoRTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m	2	6.45	1.40	1.50	27.09		cum
	Under Pipe	1	4.70	1.53	1.50	10.79		cum
						37.88	681.84	Total
2	Plain Cement Concrete 1:3:6 with B-2 40mm nominal mix in foundation 5 with crushed stone aggregate 40mm nominal size mechanically mixed, placed in foundation and compacted by vibrator including curing for 14 days. (MoRTH specification clause : 2100)	2	6.15	1.40	0.15	2.58	46.44	cum
3	Providing and laying Reinforced R-8 cement concrete pipes NP-3 for 20.4 culverts including jointing ends and fixing collar with cement mortar 1:2. (MoRTH specification : clause 1000/2900). 1000mm dia	1	7.50	-	-	7.50	135.00	RM
4	Providing and laying Plain Cement R-8 Concrete M-15 mix with stone 2 aggregate 20mm nominal size mechanically mixed and vibrated in foundation upto a depth of 1.50 below ground level and or 1.50mtr. above ground/bed level.	2	6.15	$\frac{(0.40+1.25)}{2}$	2.43	24.66		cum
	Deduction	2	$\frac{3.142}{4}$	$(1.23)^2$	$\frac{(1.01+0.60)}{2}$	1.91		
						22.75	409.50	cum.

NAME OF ROAD :-

Construction of Aamli Fala Bridge and Road

Type of cross Drainage works : Pipe Culvert - 1000mm Dia. 1Row

M³/Sec.
: HSMB

District : DUNGARPUR

1 Row 1000mm dia. 18 Width = 7.50

Sl.No. 1	Description of Item 2	No. 3	L(m) 4	B(m) 5	D(m) 6	Quantity 7	Qty 2Nos 8	Unit 9
5	Providing and laying Plain Cement R-8 Concrete M-15 Mix with stone 3 aggregate 20mm nominal size mechanically mixed and vibrated, in superstructure upto 4 meter above 1.50 m level from ground/bed level. [MoRTH Specification : Clause 1000/1700]	2	0.40	0.40	0.60	0.19	3.42	cum.

ATCH

Mahendras
[M.S. Saini]
Junior Engineer
PWD. Sub. Dn. Jhonthri

Pavan Kumawat
(Pavan Kumawat)
Assistant Engineer
PWD Sub Dn Jhonthri

DETAILED ESTIMATE

COST ESTIMATE FOR CONSTRUCTION OF CROSS DRAINAGE WORKS

(Culverts and Minor Bridges)

Construction of Aermil Fala Bridge and Road

NAME OF ROAD :-

type of cross Drainage works : Pipe Culvert - 1000mm Dia. 2Row M³/Sec.
 0 District : DUNGARPUR : HSMB

2 Row 1000mm dia. ϕ Width = 7.50

Sl.No.	Description of Item	No.	L (m)	B (m)	D (m)	Quantity	Qty 0 Nos	Unit
1	1 Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. n ordinary soil [MoRTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m	3	4	5	6	7	8	9
	Under Pipe	2	8.28	1.40	1.50	34.78		cum
		2	4.70	1.53	1.50	21.57		cum
						56.35	450.80	Cum
2	2 Plain Cement Concrete 1:3:6 with 40mm B-2 nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed, placed in foundation and compacted by vibrator including curing for 14 days. (MoRTH specification clause : 2100)	2	7.98	1.40	0.15	3.35	26.80	cum
3	3 Providing and laying Reinforced cement R-8 concrete pipes NP-3 for culverts including jointing ends and fixing collar with cement mortar 1:2. (MoRTH specification : clause 1000/2900). 1000mm dia	2	7.50	-	-	15.00	120.00	RM
4	4 Providing and laying Plain Cement R-8 Concrete M-15 mix with stone aggregate 20mm nominal size mechanically mixed and vibrated in foundation upto a depth of 1.50 below ground level and or 1.50mtr. above ground/bed level.	2	7.98	$\frac{(0.40+1.25)}{2}$	2.43	32.00		
	Deduction	4	$\frac{3.142}{4}$	$(1.23)^2$	$\frac{(1.01+0.50)}{2}$	3.83	225.36	cum.
						28.17		

NAME OF ROAD :-

Construction of Aamli Fala Bridge and Road

Type of cross Drainage works : Pipe Culvert - 1000mm Dia. 2Row

M³/Sec.

0 District : DUNGARPUR
2 Row 1000mm dia. 8 Width = 7.50

: HSMB

Sl.No.	Description of Item	No.	L (m)	B (m)	D (m)	Quantity	Qty ⁰ Nos	Unit
1		3	4	6	6	7	8	9
6	Providing and laying Plain Cement							
R-8	Concrete M-15 Mix with stone							
3	aggregate 20mm nominal size							
	mechanically mixed and vibrated, in							
	superstructure upto 4 meter above 1.50							
	m level from ground/bed level. [MoRTH							
	Specification : Clause 1000/1700]	2	0.40	0.40	0.60	0.19	1.52	cum.

AT CH.

Malashetti

[M.S. Seini]

Junior Engineer
P.W.D. Sub. En. Thonthri

Pavan Kumawat

(Pavan Kumawat)
Assistant Engineer
P.W.D Sub En Thonthri

DETAILED ESTIMATE

COST ESTIMATE FOR CONSTRUCTION OF CROSS DRAINAGE WORKS (Culverts and Minor Bridges)

Construction of Aamli Fala Bridge and Road

NAME OF ROAD :-

Type of cross Drainage works : Syphon - 500mm Dia. 1Row Design Discharge : M³/Sec.

District : DUNGARPUR Type of Soil : HSMB

PART - Syphon

Sl.No	Description of Item	No	L (m)	B (m)	D (m)	Quantity	Qty for 4Nos	Unit
1	2	3	4	5	6	8	9	9
1	Earthwork in excavation for foundation of structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. n ordinary soil [MoRTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m	2	1.90	1.90	1.00	7.22		
		1	6.80	0.70	1.00	4.76		
					Total	11.98	95.84	cum
2	Providing, laying and spreading of granular sub-base by providing coarse graded Material, spreading in uniform layers with motor grader on prepared surface mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per MoRTH Specification : clause No. 401.Grading-II	2	1.90	1.90	0.10	0.72		
		1	6.80	0.70	0.15	0.71		
		1	6.80	0.70	0.70	3.33		
			<u>3.142</u>					
	Duduct for pipe	1	4	(0.70) ²	6.80	<u>-2.62</u>		
					Total	2.14	17.14	cum
3	Providing and laying Reinforced cement concrete pipes NP-3 for culverts including jointing ends and fixing collar with cement mortar 1:2. (MoRTH specification : clause 1000/2900) 500mm dia	1	7.50	-	-	7.50	60.00	Mtr.

NAME OF ROAD :-

Construction of Aamli Fala Bridge and Road

Type of cross Drainage works : Syphon - 500mm Dia. 1Row

Design Discharge : M³/Sec.

District : DUNGARPUR

Type of Soil : HSMB

8 No Width = 7.50

PART - Syphon

Sl.No	Description of Item	No	L (m)	B (m)	D (m)	Quantity	Qty for 4Nos	Unit
1	2	3	4	5	6	8	9	9
4	Providing and laying Plain Cement							
R-8	Concrete M-15 mix with stone							
2	aggregate 20mm nominal size							
	mechanically mixed and vibrated in							
	foundation upto a depth of 1.50 below							
	ground level and or 1.50mtr. above							
	ground/bed level.	6	1.25	0.35	1.45	3.81		
		2	1.25	0.35	1.95	1.71		
			<u>3.142</u>					
	Duct for pipe	2	4	(0.70) ²	0.35	<u>-0.33</u>		
						5.19	41.48	cum
5	Providing and laying RCC M-20 mix							
R-8	with stone aggregate 20mm (nominal							
5	size) mechanically mixed and vibrated							
	for RCC work in slab in superstructure							
	upto 4 mtr. above 1.5 mtr. level from							
	Bed/Ground level, excluding steel							
	reinforcement but including centering							
	and shuttering, laid in position. (MoRTH							
	specification : Clause 1000/1700)	8	1.25	0.35	0.075	0.26	2.10	sqm
5	Plastering 20mm thick with cement							
B-3	mortar (1:3) on Masonry work in sub							
8	structure as per technical	8	0.90	-	1.45	10.44	83.52	sqm
	specifications.							

Mahendra Singh

M.S. Saini

Junior Engineer

P.W.D. Sub. Dn. Jhonthri

Pavan Kumawat

(Pavan Kumawat)

Assistant Engineer

PWD Sub Dn Jhonthri

DETAILED ESTIMATE

Type of cross Drainage works : 'Vented cause way.(1000mm Dia.)
 Vented cause way.(1000mm Dia.)

Type of Work		10 Row		M3 /Sec.		: HSMB			
S.N.	Desscription of Item	Nos	Length (M)	width (M)	Height / Thickness (m)	Quantity	Unit		
1	2	3	4	5	7	8			
1	VCW								
B-1	Earthwork in excavation for foundation of structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. n ordinary soil [MoRTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m	1	1	x	60.0	2.00	2.50	300.00	cum
1.1.4		1	1	x	60.0	2.50	3.50	525.00	cum
		1	2	x	60.0	0.75	0.75	87.50	cum
		1	4	x	60.0	3.00	0.50	360.00	cum
						Total Qty.		1252.50	cum
2	Plain Cement Concrete 1:3:6 with 40mm nominal mix in foundation with crushed stone aggregate 40mm nominal size mechanically mixed, placed in foundation and compacted by vibrator including curing for 14 days. (MoRTH specification clause : 2100)	1	1	x	60.0	2.00	0.15	18.00	cum
B-2		1	1	x	60.0	2.50	0.15	22.50	cum
5		1	1	x	60.0	6.70	0.10	40.20	cum
						Total		80.70	cum
3	Providing and laying Reinforced cement concrete pipes NP-3 for culverts including jointing ends and fixing collar with cement mortar 1:2. (MoRTH specification : clause 1000/2900). 1000mm dia	1	10	x	7.50	-	-	75.00	RM
R-8	Providing and laying Plain Cement Concrete M-15 mix with stone aggregate 20mm nominal size mechanically mixed and vibrated in foundation upto a depth of 1.50 below ground level and or 1.50mtr. above ground/bed level.	1	1	x	60.00	$\frac{2.25+0.50}{2}$	3.80	313.50	cum
		1	1	x	60.00	$\frac{1.90+0.50}{2}$	3.23	232.56	cum
		1	1	x	20.00	$\frac{1.40+0.5}{2}$	2.70	51.30	cum
	Hauching	1	1	x	30.00	6.70	0.50	100.50	Cum
						Total		697.86	cum
	Deduction	1	10	x	3.142	1.23	$\frac{6.16+0.5}{2}$	68.33 (-)	
					4			629.53	Net

Type of cross Drainage works : 'Vented cause way.(1000mm Dia.)

Vented cause way.(1000mm Dia.)

S.N.	Description of Item	1		60 M		10 Row		M3 /Sec. : HSMB	
		Nos	Length (M)	width (M)	Height / Thickness (m)	Quantity	Unit		
1	2	3	4	5	6	7	8		
6	Providing cement plaster 20mm thick in								
R-8	cement mortar 1:3 at all levels (MORTH	1	2	x	60.00	-	1.20	144.00	sqm.
19	specification : clause 1000/1300)	4	1.01	$\frac{+}{2}$	0.40	-	1.20	5.81	sqm.
	Deduction	1	10	x	$\frac{3,142}{4}$	1.23	1.23	11.88	(-)
								161.69	Net
7	Providing and laying graded stone								
B-5	aggregate filter Material underneath								
4	pitching on slopes complete as per								
	drawing and technical specification.								
	(MORTH specification : clause 2504).								
		1	60.0	6.24	0.30			112.32	cum
		1	30.0	3.00	0.30			27.00	cum
								139.32	Net
8	Providing and laying Plain Cement								
R-8	Concrete M-15 Mix with stone aggregate								
3	20mm nominal size mechanically mixed								
	and vibrated, in superstructure upto 4								
	meter above 1.50 m level from								
	ground/bed level. [MoRTH Specification :								
	Clause 1000/1700]	1	3	x	1.73	6.68	0.20	6.93	cum.
9	Providing and laying RCC M-20 mix with								
R-8	stone aggregate 20mm (nominal size)								
4	mechanically mixed and vibrated for RCC								
	work in slab in superstructure upto 4 mtr.								
	above 1.5 mtr. level from Bed/Ground								
	level, excluding steel reinforcement but								
	including centering and shuttering, laid in								
	position. (MoRTH specification : Clause								
	1000/1700)								
		1	1	x	60.00	6.70	0.15	60.30	cum
	on wall	1	2	x	60.00	0.40	0.25	6.00	cum
	Toe wall	1	2	x	60.00	0.45	0.075	2.03	cum
		1	1	x	30.00	$\frac{0.30+0.45}{2}$	0.45	5.06	cum
	Gurd Stone								
								73.39	cum
10	Steel reinforcement in substructure								
B-3	complete. [MoRTH Specification : Section								
9.1	1600] HYSD reinforcement	@	60.00 Kg/cum	x	73.39			4403.25	kg
							or	4.40	Tonne
11	Providing A.C. pipe Weep holes 150mm								
B-3	dia in Masonry/plain Concrete/Reinforced								
16	Concrete abutment, wing wall/return wall								
	complete as per drawing and technical								
	specification. [MoRTH Specification :								
	Clause 2205].	1	60	60.00				3600.00	Each

Mahendra H

M.S. Saini

Junior Engineer

P.W.D. Sub Dn Thonthori

Pavan Kumawat

(Pavan Kumawat)

Assistant Engineer

PWD Sub Dn Antri

Thonthori

Construction of Aarmil Fala Bridge and Road

Widening of (RCC Slab Culvert)

10 M.-

3 SPAN

DETAILS

S.No.	Description of Item.	No.	L	W	H/D	Quantity
1	Earthwork in excavation for foundation of structure complete with all lift, dewatering, B-1 shoring and shuttering etc. including 1.1.4 refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification : Clause 304]. In ordinary soil [MORTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m	2	8.50	8.80	1.20	179.52 cum
	Abutment	2	5.90	13.10	1.20	185.50 cum
	Piers	1	60.00	3.50	1.20	252.00 cum
	For Wall				Total =	617.02 cum
2	Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering, 1.3.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification : Clause 304]. In Ordinary/Soft rock (not requiring blasting) [MORTH Specification : Clause 301] BY MECHANICAL MEANS	2	8.50	8.80	1.50	224.40 cum
	Abutment	2	5.90	13.10	1.50	231.87 cum
	Piers	1	60.00	3.50	1.00	210.00 cum
	For Wall				Total =	666.27 cum
3	Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering, 1.4.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification : Clause 304]. In Hard Rock [MORTH Specification : Clause 301, 302, 303]. Blasting Prohibited	2	8.50	8.80	1.50	224.40 cum
	Abutment	2	5.90	13.10	1.50	231.87 cum
	Piers				Total =	456.27 cum